

Mini project 2

Name: Prathiksha Harish Baleri

USN:4CB22CB044

Domain: data science

1. Title of the Project

Customer Segmentation using K-Means Clustering Algorithm

2. Objective

The main objectives of this project are:

- **Group customers** based on similar characteristics.
 - **Identify different types** of customers and their behaviors.
 - **Help businesses improve** marketing strategies through personalization.
 - **Increase sales** and improve overall customer satisfaction.
-

3. Problem Statement

Businesses often have large numbers of customers, but they face several challenges:

- **Mass Marketing:** Treating all customers the same regardless of individual needs.
- **Identification Gap:** Inability to identify high-value or "VIP" customers.
- **Resource Waste:** Wasting marketing resources on non-responsive segments.
- **Churn:** Failure to recognize and retain at-risk customers.

This project aims to solve these issues by segmenting customers into distinct groups using machine learning clustering techniques.

4. Proposed Solution

Develop a data-driven system that:

- **Processes Data:** Takes customer attributes like annual income, spending score, age, and gender.
 - **Applies Algorithms:** Utilizes the K-Means clustering algorithm to find natural groupings.
 - **Categorizes Segments:** Divides customers into meaningful groups, such as:
 - **High Spenders:** High income, high spending.
 - **Budget Customers:** Low income, low spending.
 - **Premium Customers:** High income, moderate spending.
 - **Potential Loyalists:** Moderate income, high spending.
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5. Technologies Used

- **Programming Language:** Python
- **Libraries:**
 - **Pandas:** Data manipulation and analysis.
 - **NumPy:** Mathematical and numerical operations.
 - **Matplotlib / Seaborn:** Data visualization and cluster plotting.
 - **Scikit-learn:** Implementation of the K-Means algorithm.
- **Tools:**
 - **Jupyter Notebook / Google Colab:** Development environment.
 - **Excel:** Dataset storage and handling.

6. Dataset Description

The project utilizes a structured dataset containing demographic and behavioral data. Typical dataset features include:

Feature Name	Description
Customer_ID	A unique identification number assigned to each individual customer.
Age	The age of the customer (used to identify generational trends).
Genre	Male/female
Annual_Income	The yearly earnings of the customer (expressed in local currency or USD).
Spending_Score (1–100)	A score assigned by the business based on customer behavior and purchasing nature.

7. System Architecture Flow:

1. Data Collection
2. Data Preprocessing
3. Feature Selection
4. Clustering Algorithm
5. Model Training
6. Cluster Visualization
7. Analysis & Interpretation

8.Methodology (Step-by-Step)

Step 1: Data Collection

- Collect customer data from:
 - o Retail stores
 - o E-commerce platforms
 - o CSV/Excel datasets

▼ Customer Segmentation using K-Means Clustering Algorithm

```
[113]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Data collection using kaggle dataset Mall customers

data input module

```
[114]: df=pd.read_csv('Mall_Customers.csv')
```

```
[115]: df.head()
```

```
[115]:
```

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
[116]: df.shape
```

```
[116]: (200, 5)
```

Step 2: Data Preprocessing

- Handle missing values
- Encode categorical data (e.g., Gender → 0/1)
- Normalize/scale data

▼ data preprocessing

data preprocessing module

```
[117]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                      Non-Null Count  Dtype
---  ---
0   CustomerID                  200 non-null    int64
1   Genre                       200 non-null    object
2   Age                         200 non-null    int64
3   Annual Income (k$)          200 non-null    int64
4   Spending Score (1-100)      200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

Step 3: Feature Selection

Select relevant features such as:

- Annual Income • Spending Score • Age

feature extraction

```
[118]: #Creating a seperate dataframe X
X=df.iloc[:,[3,4]].values
#retrieve data
#iloc used to extract data based on columns
X #The 3rd and 4th columns r shown
```

```
[118]: array([[ 15,  39],
        [ 15,  81],
        [ 16,   6],
        [ 16,  77],
        [ 17,  40],
        [ 17,  76],
        [ 18,   6],
        [ 18,  94],
        [ 19,   3],
        [ 19,  72],
        [ 19,  14],
        [ 19,  99],
        [ 20,  15],
        [ 20,  77],
        [ 20,  13],
        [ 20,  79],
        [ 21,  35],
        [ 21,  66],
        [ 23,  29],
        [ 23,  98],
        [ 24,  35],
        [ 24,  73],
        [ 25,   5],
        [ 25,  73],
        [ 28,  14],
        [ 28,  82],
        [ 28,  32],
        [ 28,  61],
        [ 29,  31],
```

Step 4: Finding Optimal Clusters (Elbow Method)

Use the Elbow Method to find the best value of K (number of clusters):

- Performing elbow method to find the optimal number of customers

using K-Means to iterate from 1 to 11 clusters plotting a Elbow plot and deciding the optimal number of clusters to be used

cluster module

```
[119]: from sklearn.cluster import KMeans
      wcss=[] #Within clusters sum of squares for k
      #this is distance between each point and the centroid in a cluster

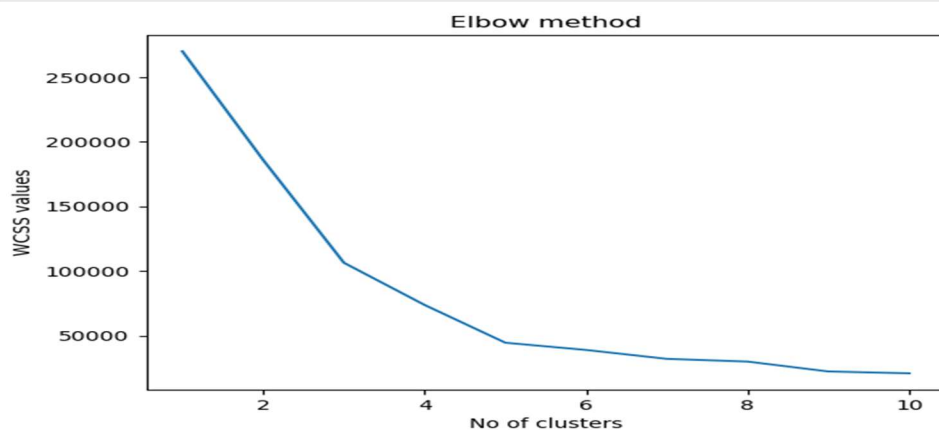
[120]: # as no of clusters increase the wcss value decrease
      #wcss value is largest when k=1

[121]: for i in range(1,11):
      kmeans=KMeans(n_clusters=i,init='k-means++',random_state=0)
      kmeans.fit(X)
      wcss.append(kmeans.inertia_)

      #kmeans.inertia is used to segregate the data points into clusters

C:\Users\HP\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when
ere are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.
  warnings.warn(
C:\Users\HP\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when
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ere are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.
  warnings.warn(

[122]: plt.plot(range(1,11),wcss)
      plt.title('Elbow method')
      plt.xlabel('No of clusters')
      plt.ylabel('WCSS values')
      plt.show()
```



we are getting elbow at k=5

so number of optimal clusters=5

Step 5: Apply K-Means Clustering Algorithm

1. Select K (number of clusters)
2. Initialize centroids
3. Assign data points to nearest centroid
4. Update centroids
5. Repeat until convergence

Initialise our KMeans model with selected optimal number of clusters(k=5)

```
[123]: kmeansmodel = KMeans(n_clusters=5, init='k-means++', random_state=42)
```

Step 6: Model Training

- Train K-Means model using dataset
- Assign cluster labels to each customer

Model prediction

```
[124]: y_kmeans = kmeansmodel.fit_predict(X)
```

```
C:\Users\HP\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: There are less chunks than available threads. You can avoid it by setting the 'n_jobs' parameter to a smaller value.
  warnings.warn(
```

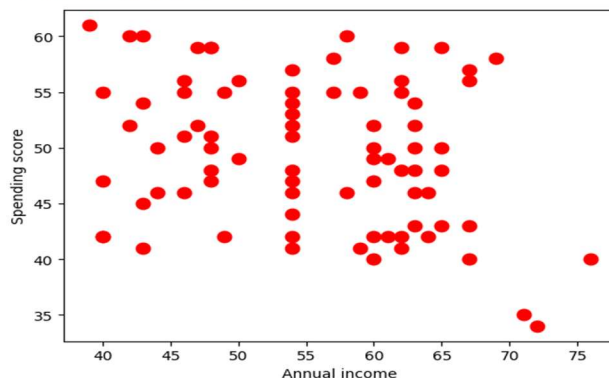
Step 7: Visualization

- Plot clusters using graphs: o Income vs Spending Score
- Different colors represent different clusters

visualization module

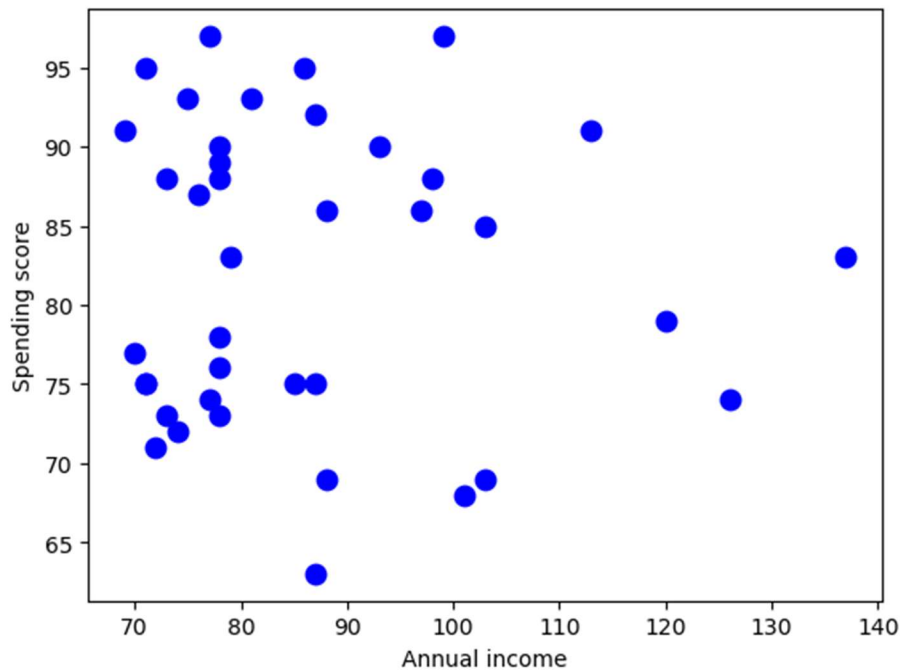
```
[125]: plt.scatter(X[y_kmeans==0,0],X[y_kmeans==0,1],s=80,c="red",label='Customer1')
plt.xlabel("Annual income")
plt.ylabel("Spending score")
```

```
[125]: Text(0, 0.5, 'Spending score')
```



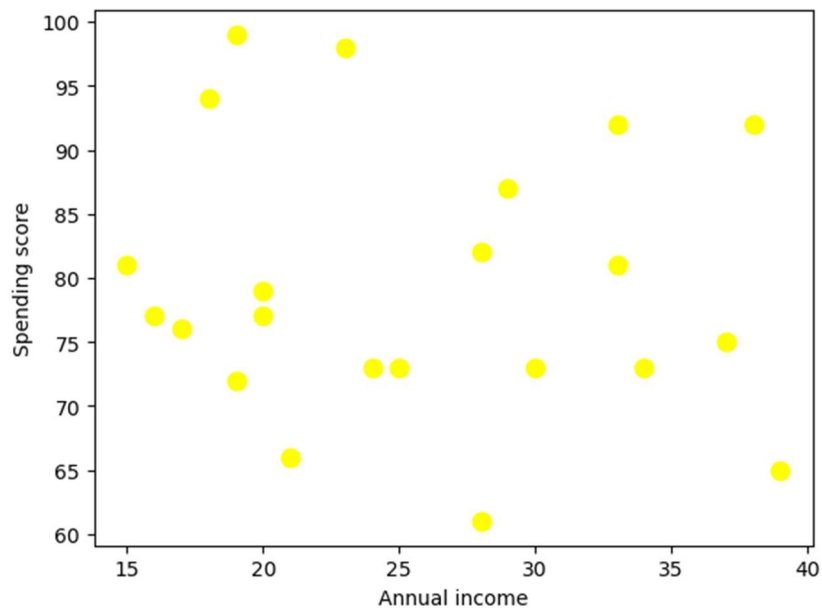
```
[126]: plt.scatter(X[y_kmeans==1,0],X[y_kmeans==1,1],s=80,c="blue",label='Customer2')
plt.xlabel("Annual income")
plt.ylabel("Spending score")
```

```
[126]: Text(0, 0.5, 'Spending score')
```



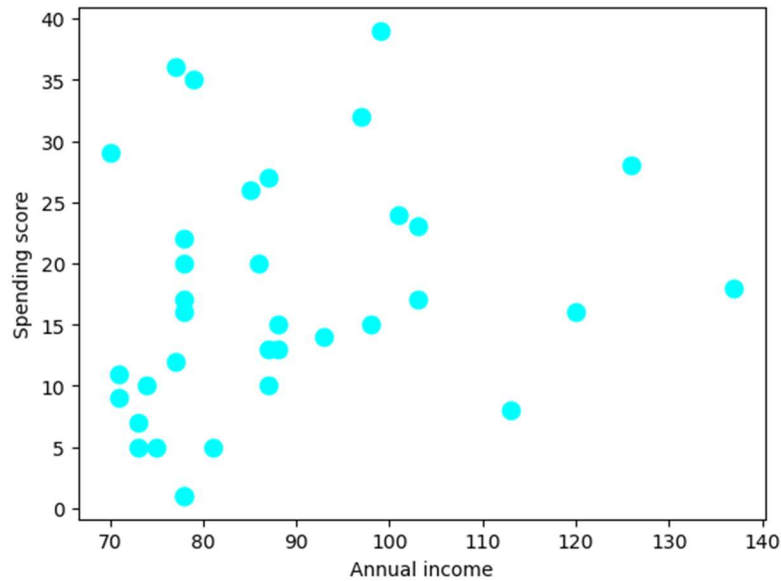
```
[127]: plt.scatter(X[y_kmeans==2,0],X[y_kmeans==2,1],s=80,c="yellow",label='Customer3')
plt.xlabel("Annual income")
plt.ylabel("Spending score")
```

```
[127]: Text(0, 0.5, 'Spending score')
```



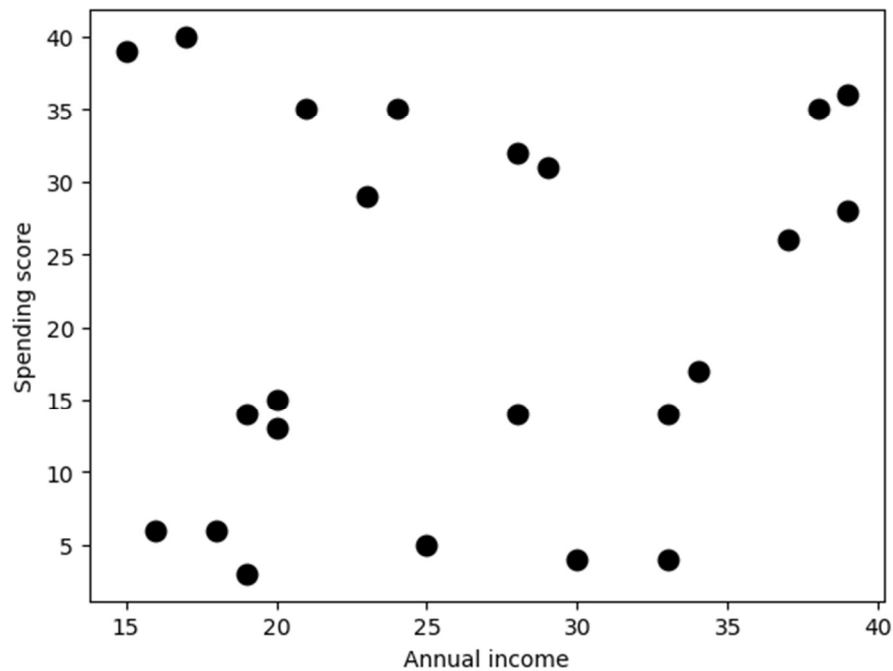
```
[128]: plt.scatter(X[y_kmeans==3,0],X[y_kmeans==3,1],s=80,c="cyan",label='Customer4')
plt.xlabel("Annual income")
plt.ylabel("Spending score")
```

```
[128]: Text(0, 0.5, 'Spending score')
```

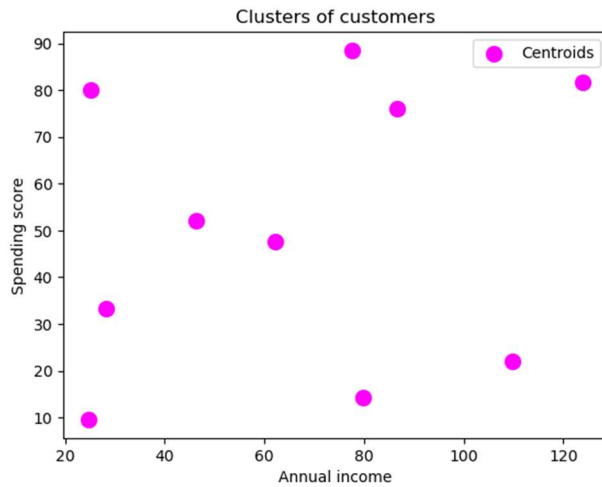


```
[129]: plt.scatter(X[y_kmeans==4,0],X[y_kmeans==4,1],s=80,c="black",label='Customer5')
plt.xlabel("Annual income")
plt.ylabel("Spending score")
```

```
[129]: Text(0, 0.5, 'Spending score')
```



```
[130]: plt.scatter(kmeans.cluster_centers[:,0],kmeans.cluster_centers[:,1],s=100,c="magenta",label="Centroids")
plt.title("Clusters of customers")
plt.xlabel("Annual income")
plt.ylabel("Spending score")
plt.legend()
plt.show()
```



```
[131]: plt.scatter(X[y_kmeans==0],X[y_kmeans==0,1],s=80,c="red",label='Customer1')
plt.scatter(X[y_kmeans==1],X[y_kmeans==1,1],s=80,c="blue",label='Customer2')
plt.scatter(X[y_kmeans==2],X[y_kmeans==2,1],s=80,c="yellow",label='Customer3')
plt.scatter(X[y_kmeans==3],X[y_kmeans==3,1],s=80,c="cyan",label='Customer4')
plt.scatter(X[y_kmeans==4],X[y_kmeans==4,1],s=80,c="black",label='Customer5')
plt.scatter(kmeans.cluster_centers[:,0],kmeans.cluster_centers[:,1],s=100,c="magenta",label="Centroids")
plt.title("Clusters of customers")
plt.xlabel("Annual income")
plt.ylabel("Spending score")
plt.legend()
plt.show()
```



Step 8: Interpretation

▼ evaluation

Top-Left Cluster: Represents customers with relatively lower annual income but a high spending score. Black-colored Cluster: Represents customers with a high annual income but a lower spending score. The third cluster stands out as the most valuable, as these individuals exhibit both decent income and a good spending score. Companies often prioritize this group to promote products effectively

Specifically: The business should reward Customer 2 (Blue) with loyalty programs to ensure retention. The business should target Customer 4 (Cyan) with specific promotions or discounts to encourage them to spend more of their high income. The business can minimize marketing costs on Customer 5 (Black) or offer them extreme budget-friendly deals.

9. System Modules

1. Data Input Module
2. Data Processing Module
3. Clustering Module
4. Visualization Module
5. Analysis Module

10. Expected Output

- Customers grouped into clusters
- Visual graphs showing segmentation
- Business insights for each group

11. Advantages

- Better customer understanding
- Targeted marketing strategies
- Increased profit and efficiency

- Personalized services
-

12. Limitations

- Requires proper selection of K
 - Sensitive to outliers
 - Results depend on data quality
-

13. Future Enhancements

- Use advanced clustering (DBSCAN, Hierarchical)
 - Real-time segmentation
 - Integration with CRM systems
 - AI-based recommendation systems
-

14. Conclusion

Customer segmentation helps businesses:

- Identify valuable customers
- Improve marketing strategies
- Make data-driven decisions

Clustering techniques like K-Means provide efficient and scalable solutions.
